Sparse Counterfactual Attribution of Crop-Yield Anomalies to Extreme-Weather Drivers: A United States Study (1990–2025)

Tran Dai Phong Lam*[1], Thu Le[1], Nguyen Quoc Hung[1], and Nguyen Trung Trinh[1]

[1]FPT University, Ho Chi Minh City, Vietnam

Abstract

Most crop-yield machine-learning studies optimize average predictive accuracy, yet the years that matter most for food security and risk management are the abnormal ones: regional crop failures. Average-accuracy models rarely say *why* a particular crop-region-year collapsed, or *which* weather extreme was responsible. We propose Sparse Counterfactual Anomaly Attribution (SCAA), an event-level explanation method that detects low-yield anomalies and attributes each one to a small, feasible set of extreme-weather drivers. Daily NASA POWER weather (1989–2025), harmonized across twelve U.S. states and aggregated over hemisphere-correct, crop-specific growing-season windows, is merged with USDA NASS yields for Barley, Canola, Oats and Wheat. A forward-time gradient-tree model predicts yield with a test R^2 of 0.809 (2016–2021) and 0.808 on an extended 2019–2025 horizon. Each crop-region yield series is detrended to separate technological drift from weather-driven variation; years whose detrended residual falls below -1 standard deviation are flagged as anomalies (215 of 1,227 observations, 17.5%). For each anomaly, a sparse, feasibility-constrained counterfactual computes the minimal weather change that would restore predicted yield to its trend, yielding (i) a dominant driver and (ii) a weather-recoverable fraction of the yield gap. Across anomalies the median recoverable fraction is 0.86, and 89% are classified as weather-driven; the remaining 11% are flagged as not weather-explainable, a useful signal for non-meteorological causes. Attribution is validated against known events: the 2012 drought anomalies are attributed to heat and growing-degree-day extremes, and the severe 2021–2022 anomalies to dry-spell and moisture extremes; anomaly counts peak in the documented drought years 2002, 2011, 2021 and 2022. SCAA turns a black-box yield model into an interpretable, event-level attribution tool. Results are predictive associations, not causal effects.

Keywords: crop yield; extreme weather; counterfactual explanation; anomaly detection; interpretable machine learning; United States.

1 Introduction

Crop-yield prediction is dominated by models that minimize average error over all crop-region-years [3, 9]. For planning and risk management, however, the decisive years are the abnormal ones: the regional failures that strain supply and reveal climate vulnerability. An average-accuracy model

*Corresponding author: phonglam2599@gmail.com

rarely answers the two questions a planner actually asks about such a year: *why* did this crop-region collapse, and *which* weather extreme was responsible? Global feature-importance scores describe the model in aggregate, not the individual event.

This study reframes weather–yield interpretation as *event-level attribution*. We propose Sparse Counterfactual Anomaly Attribution (SCAA): detect low-yield anomalies, then attribute each to a small, feasible set of extreme-weather drivers using a sparse counterfactual. The method answers, for a specific bad year, "the smallest weather change that would have made this year normal," and identifies the weather indicators that change. The approach is organized around three questions:

- **RQ1.** After separating long-run trend from year-to-year variation, which crop-region-years are genuine low-yield anomalies?
- **RQ2.** What fraction of each anomaly is recoverable by feasible weather change, and which extreme-weather indicator is its dominant driver?
- **RQ3.** Do the attributions agree with documented drought and heat events?

The contribution is a method that ports sparse counterfactual explanation [8] from instance-level classification to event-level yield-anomaly attribution, with a continuous weather-recoverable fraction that also flags anomalies that weather cannot explain. We study Barley, Canola, Oats and Wheat across twelve U.S. states, 1990–2025, on a source-harmonized daily-weather record. All results are predictive associations and planning signals, not causal effects.

2 Related Work

Machine learning for yield prediction has been reviewed extensively, with weather, soil and historical yield as common inputs across classical and deep models [3, 6, 9]; because no single model dominates across crops and regions, comparative studies are standard [2, 5]. Interpretability work warns that black-box yield models can mislead climate and planning conclusions without careful, domain-aware explanation [1, 7]. Counterfactual explanation—minimal, feasible input changes that alter a prediction—is a standard post-hoc tool [8]; we adapt its feasibility and sparsity principles from instance classification to yield-anomaly attribution. Climate-attribution studies compare observed outcomes against a counterfactual without climate trends [4]; SCAA applies an analogous logic at the level of individual crop-region-year anomalies, while emphasizing that attributions are model-internal and not causal.

3 Data

The unit of analysis is a crop-region-year; the target is yield in t/ha. We study the overlap winter and spring crops Barley, Canola, Oats and Wheat across twelve U.S. states (Colorado, Illinois, Iowa, Kansas, Minnesota, Montana, Nebraska, North Dakota, Oklahoma, South Dakota, Texas, Washington). Yields are from USDA NASS Quick Stats, converted from bushels (or pounds, for canola) per acre to t/ha with crop-specific test weights. Daily weather (maximum and minimum temperature, corrected precipitation, all-sky surface radiation) is from NASA POWER, used uniformly for all states so that no cross-source measurement difference confounds the analysis. The merged frame contains 1,257 crop-region-year observations spanning 1990–2025.

4 Method

4.1 Hemisphere-correct growing-season features

Daily records are aggregated into interpretable growing-season indicators (Figure 1): seasonal rainfall totals and means, dry-day counts and maximum dry-spell length, long dry-spell event counts, heavy-rain day counts and rolling 1/3/7-day maxima, heat-day counts and consecutive-heat runs, heatwave events, heat- and growing-degree days, hot-dry days, frost and cold days, frost events, radiation summaries, and early/mid/late thirds of the window. Crucially, the aggregation window is crop- and latitude-specific rather than a single fixed window: U.S. winter wheat (southern and Pacific-Northwest states) is aggregated over September–June, while spring crops (barley, oats, canola, and northern-plains spring wheat) are aggregated over April–September. This phenology-correct windowing captures the over-wintering period of winter wheat (e.g. the establishment-stage frost exposure) that a fixed warm-season window omits.

4.2 Detrending and anomaly definition

Long-run yield change is dominated by technological and managerial trend rather than by weather, so we separate the two before attribution. For each crop-region series we fit an ordinary-least-squares trend $\hat{y}_{c,r}(t) = \beta_0 + \beta_1 t$ and form the residual $e_{c,r,t} = y_{c,r,t} - \hat{y}_{c,r}(t)$, standardized within series to $z_{c,r,t} = e_{c,r,t}/\sigma_e$. A crop-region-year is a *low-yield anomaly* if $z_{c,r,t} < -1$. This removes the trend that would otherwise dominate any attribution and isolates the weather-driven shortfall.

4.3 Yield model

An Extremely Randomized Trees regressor (400 trees, minimum leaf size 2) maps the growing-season features, the year index, location and crop/region identifiers to yield. Numeric features are median-imputed and standardized; categorical features are one-hot encoded. The model is fit under a forward-time protocol (train ≤ 2015, test ≥ 2016).

4.4 Sparse counterfactual anomaly attribution (SCAA)

Let $\hat{f}$ be the yield model and, for an anomaly (c, r, t), let $\tau = \hat{y}_{c,r}(t)$ be the trend-expected "normal" yield. With $\mathbf{w}$ the observed weather vector, SCAA seeks the minimal feasible weather change that raises the predicted yield toward τ:

$$\boldsymbol{\delta}^{\star} = \arg\min_{\boldsymbol{\delta}} \ \|\boldsymbol{\delta} \oslash \boldsymbol{\sigma}_r\|_1 \quad \text{s.t.} \quad \hat{f}(\mathbf{w} + \boldsymbol{\delta}) \ \uparrow \ \tau, \ \ \mathbf{w} + \boldsymbol{\delta} \in \mathcal{W}_r, \tag{1}$$

where $\boldsymbol{\sigma}_r$ is the per-feature regional standard deviation and $\mathcal{W}_r$ the region's observed feasible box (preventing out-of-distribution counterfactuals). Because $\hat{f}$ is a tree ensemble, we solve by greedy coordinate search over interpretable indicators with a sparsity budget. Rather than a binary repaired/not, we report the continuous *weather-recoverable fraction*

$$\phi_{c,r,t} = \frac{\hat{f}(\mathbf{w} + \boldsymbol{\delta}^{\star}) - \hat{f}(\mathbf{w})}{\tau - \hat{f}(\mathbf{w})}, \tag{2}$$

the share of the modelled yield gap that feasible weather change can recover, and the *dominant driver*, the indicator with the largest standardized change in $\boldsymbol{\delta}^{\star}$. An anomaly is *weather-driven* if $\phi \geq 0.5$; a low ϕ flags an anomaly that weather alone cannot explain. All attributions are post-hoc and do not influence model fitting.

5 Results

5.1 Predictive performance

On harmonized data with phenology-correct windows the yield model attains a test R^2 of 0.809 (RMSE 0.525 t/ha) over 2016–2021, and 0.808 (RMSE 0.536) over an extended 2019–2025 horizon, confirming stable forward-time skill into recent years (Figure 2). Predictability is markedly regime-dependent: spring crops reach $R^2 = 0.842$ while winter wheat, despite its correct September–June window, remains harder at $R^2 = 0.543$, reflecting the larger weather-driven variance of semi-arid winter-wheat regions.

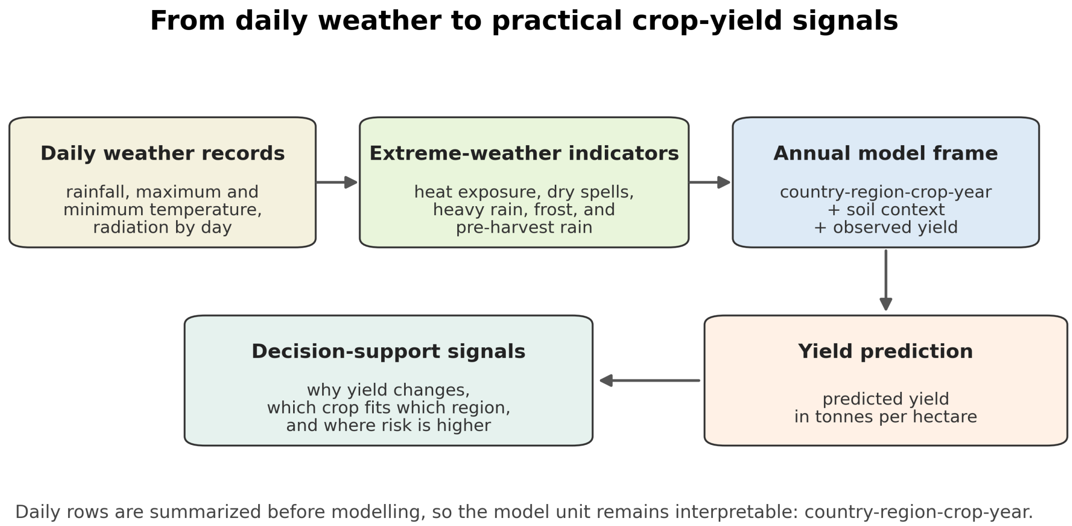

Figure 1: From daily weather to interpretable growing-season indicators. Daily records are aggregated over crop- and latitude-specific windows into extreme-weather features used for prediction and anomaly attribution.

5.2 Anomaly detection

After detrending, 215 of 1,227 crop-region-years (17.5%) are flagged as low-yield anomalies ($z < -1$). Their temporal distribution (Figure 3) concentrates in documented U.S. drought years: anomaly counts peak in 2002, 2011, 2021 and 2022, with 2012 also prominent—each a recognized drought or heat year on the Plains. This temporal alignment is independent corroboration that the detrended anomalies capture genuine weather-driven failures rather than statistical noise.

5.3 Counterfactual attribution

SCAA assigns each anomaly a weather-recoverable fraction and a dominant driver (Figure 4). The median recoverable fraction is 0.86, and 89% of anomalies are weather-driven ($\phi \geq 0.5$); the remaining 11% have low ϕ and are flagged as not weather-explainable, a useful pointer to non-meteorological causes (pests, management, market-driven abandonment, or moisture excess beyond the model's sensitivity). The dominant drivers form an agronomically coherent set (Table 1): long dry-spell events and frost events lead, followed by seasonal rainfall deficits, heat-day exposure, excess-wet

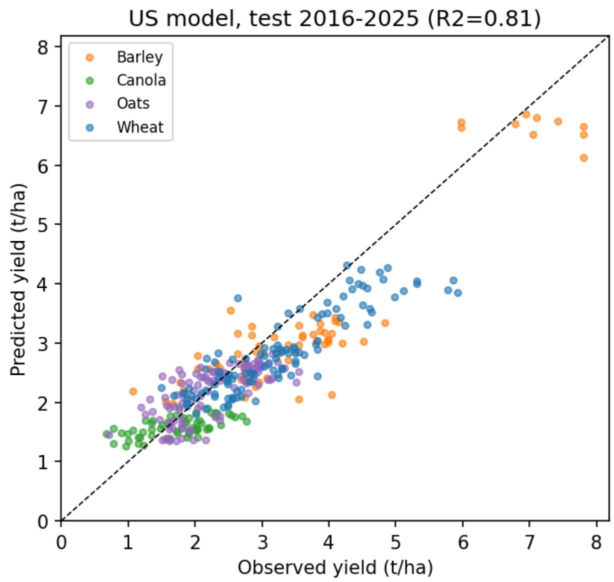

Figure 2: Observed versus predicted yield, forward-time test 2016–2025. Calibration is close to the identity line across crops.

days and radiation, reflecting that U.S. crop failures arise from both drought/heat and, less often, cold and excess-moisture extremes.

Table 1: Most frequent dominant weather driver across weather-driven anomalies.

Dominant driver	# anomalies
Long dry-spell events ($\geq$14 days)	23
Frost events ($\geq$2 consecutive days)	18
Seasonal rainfall (deficit)	16
Heat days ($T_{\max} \geq 30°$C)	15
Wet days (excess)	12
Seasonal radiation	11
Mean maximum temperature	9
Heatwave events (3-day, 30°C)	9

5.4 Event validation

The attributions agree with known events. In the 2012 Plains drought, the Kansas Oats anomaly is attributed to growing-degree-day and heat extremes (recoverable fraction 0.93), North Dakota Canola to consecutive heat days (0.86), and South Dakota Barley to heatwave events (0.60)—all heat-driven, consistent with the 2012 heat-drought. The most severe recent anomalies, in 2021–2022, are attributed predominantly to dry-spell and dry-day extremes (e.g. Oklahoma Oats 2022 and South Dakota Barley 2021), matching the 2021 Northern-Plains/Pacific-Northwest drought and the 2022 droughts. A minority of 2021 North Dakota anomalies have low recoverable fraction with excess-moisture drivers, correctly flagged by SCAA as not cleanly weather-explainable.

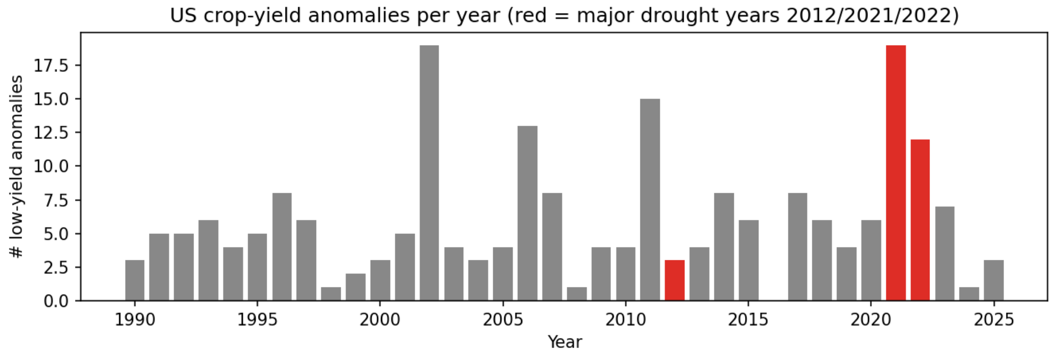

Figure 3: Low-yield anomalies per year; major drought years (2012, 2021, 2022) highlighted. Anomaly frequency tracks documented drought episodes.

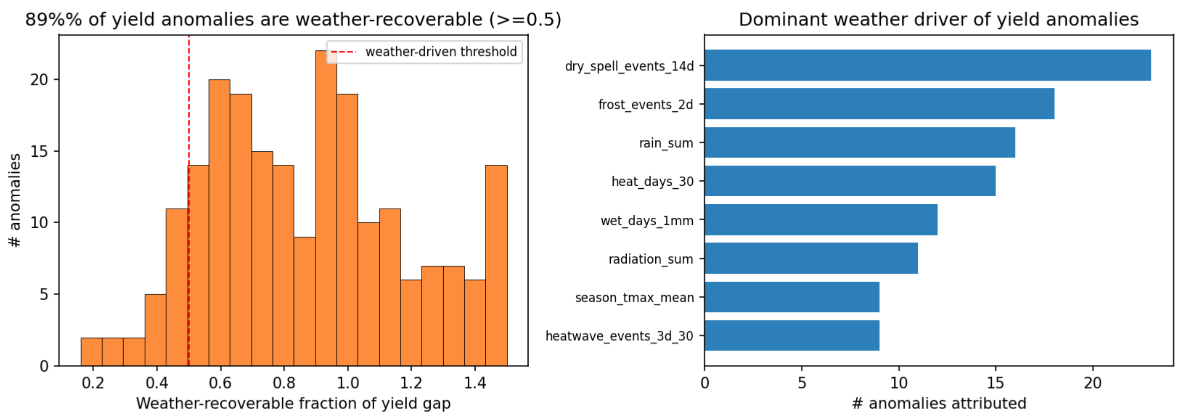

Figure 4: SCAA outputs. Left: distribution of the weather-recoverable fraction; 89% of anomalies exceed the 0.5 weather-driven threshold (median 0.86). Right: frequency of the dominant weather driver across attributed anomalies.

6 Discussion

SCAA converts a competitive but opaque yield model into an interpretable, event-level tool. The two-stage design—detrend, then attribute the residual anomaly—matters: long-run yield change is trend-dominated, so attributing raw change to weather would be misleading, echoing cautions in explainable yield modelling [1] and climate-trend attribution [4]. The continuous weather-recoverable fraction is more honest than a binary explanation: it quantifies how much of each failure weather can account for and, by flagging low-ϕ anomalies, distinguishes weather failures from those requiring non-weather explanations. Validation against 2012 and 2021–2022 shows the attributions recover the correct physical driver for documented events.

7 Limitations and Future Work

Attributions are model-internal predictive associations, not causal effects, and depend on the model's weather sensitivity; the greedy counterfactual margin is a conservative lower bound. State-level aggregation cannot resolve within-state management, irrigation, cultivar or soil variation. Linear detrending may under- or over-fit non-linear technology curves for some series. Future work should add irrigation and management covariates, finer spatial resolution, bootstrap intervals on the recoverable fraction, and a synthetic benchmark with known weather/trend contributions to validate the attribution itself.

8 Conclusion

We presented SCAA, a sparse-counterfactual method that detects low-yield anomalies and attributes each to its dominant extreme-weather driver with a weather-recoverable fraction. On twelve U.S. states and four crops (1990–2025), a forward-time model reaches $R^2 = 0.81$, anomalies align with documented droughts, 89% of anomalies are weather-driven, and event-level attributions recover the correct drivers for the 2012 and 2021–2022 events while honestly flagging the cases weather cannot explain. SCAA makes extreme-weather yield interpretation operational at the level of the individual events that matter most for risk.

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